

I can't generate a PDF file directly. However, I can provide a complete, copy-paste-ready response that you can place into Word and export as PDF.

Below are the answers as the customer representative of **Lanka Earth Movers (Pvt) Ltd** for your **Web-based JCB Management System (DozerNet)**.

DozerNet – Client Responses to Clarification Questions

Client: Lanka Earth Movers (Pvt) Ltd

Representative: Mr. Nuwan Perera, Operations Manager

Q1. Automated Validation

Manual admin review alone is not sufficient. The system should automatically validate required fields, prevent duplicate records, check booking conflicts, and verify machine availability before approval.

Q2. Document Verification

Manual admin verification of NIC and driving licence documents is sufficient for this project. No government API integration is required.

Q3. Machine Availability Checks

The system should automatically block bookings if:

- The machine already has an approved booking.
- The machine is under maintenance.
- The machine has been marked unavailable by an admin.

Q4. Customer Contact Validation

If notifications fail, the system should flag the account and prompt the customer to update contact details. Future bookings should still be allowed, but admins should see the warning.

Pricing Model

Q5. Pricing Rules

The final pricing model should include:

- Machine-specific daily rates.
- Minimum booking period of 1 day.
- District-based transportation surcharge.
- No seasonal pricing for this version.

Example:

- JCB Backhoe: Rs. 18,000/day
 - Excavator: Rs. 25,000/day
 - Transportation: Rs. 2,000 to Rs. 10,000 depending on district.
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Cancellation & Refund Policy

Q6. Cancellation Rules

Pending bookings:

- Can be cancelled at any time with no charges.

Approved but unpaid bookings:

- Can be cancelled before the start date.

Paid bookings:

- More than 48 hours before hire date: Full refund.
 - Less than 48 hours before hire date: 20% cancellation charge.
 - After hire start date: No refund.
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Payment Simulation

Q7. Demo Payment Validation

The system should perform basic validation:

- Card number format check.
- Expiry date validation.
- CVV must be 3 digits.

Actual bank integration is not required.

Q8. Partial Payments

Partial payments should be supported.

The system should:

- Record paid amount.
 - Display remaining balance.
 - Send reminders until the balance is cleared.
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Booking Conflict Rules

Q9. Booking Conflicts

A customer may request multiple machines in one booking.

A machine cannot be double-booked during overlapping dates.

Conflict checking should be based on booking start and end dates, not only by full day.

Project Documentation

Q10. Design Document

Yes, the Design Document is required.

It should include:

- Use Case Diagram
- Class Diagram
- Activity Diagrams
- Ethical Considerations
- Agile Sprint Summary

This document should be completed before the final evaluation.

Fleet & Categories

Q11. Machine Categories

Customers should browse machines by:

- JCB Backhoe Loaders
- Excavators
- Bulldozers
- Wheel Loaders
- Skid Steer Loaders
- Motor Graders
- Rollers/Compactors
- Dump Trucks

Q12. Machine Photos

Photos are required for all listings.

Both:

- Admins
- Private machine owners

can upload photos.

Locations

Q13. District Coverage

The system should support all 25 districts of Sri Lanka.

Identity & Verification

Q14. NIC Formats

Both NIC formats should be accepted:

Old:

- 123456789V
- 123456789X

New:

- 200012345678
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Operators

Q15. Operator Assignment

Operator assignment should be a manual admin decision.

The system may suggest available operators, but final selection should be made by an admin.

Q16. Driver Payments

For this version, operator payments are handled outside the platform.

The system only records assignments.

Private Owners

Q17. Commission Model

Yes.

The company charges a commission of 10% from bookings completed using private-owner machines.

The system should track this commission.

Maintenance

Q18. Maintenance Scheduling

Admins should be able to manually schedule maintenance.

The system should also support interval-based reminders:

- Every 3 months
 - After a specific operating-hour threshold
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Notifications

Q19. Notification Method

In-app notifications are required.

A realistic demo email log should also be available showing:

- Booking approvals
- Payment confirmations
- Maintenance alerts
- Booking cancellations

Reporting

Q20. Required Reports

The system should provide:

1. Revenue by month.
2. Revenue by machine type.
3. Booking counts.
4. Fleet utilization report.
5. Maintenance report.
6. Customer activity report.
7. Outstanding payment report.

Reports should:

- Be viewable on screen.
 - Be exportable to CSV and PDF.
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Accounts

Q21. Multiple Roles

One person should be able to have multiple roles under a single account.

For example:

- Customer + Private Owner
 - Customer + Operator
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Invoicing

Q22. Tax Handling

Invoices should include VAT.

Example:

- Rental Fee
 - Transport Fee
 - VAT (18%)
 - Total Amount
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Localization & Accessibility

Q23. Language Support

English is sufficient for the initial release.

Future versions may include:

- Sinhala
- Tamil

Q24. Accessibility

No formal WCAG certification is required.

The website should simply:

- Use readable fonts.
 - Provide clear navigation.
 - Be mobile-friendly.
 - Support basic accessibility practices.
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Security

Q25. Security Policies

Requirements:

- BCrypt password hashing.
 - Minimum 8-character password.
 - At least one uppercase letter.
 - At least one number.
 - Session timeout after 30 minutes of inactivity.
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Booking Window

Q26. Advance Booking Rules

- Same-day bookings are allowed.
 - Maximum booking window: 180 days in advance.
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Invoice Numbering

Q27. Invoice Format

Use a structured numbering format:

INV-YYYY-00001

Example:

INV-2026-00001

INV-2026-00002

Numbers should automatically increment.

Additional Business Requirements

To make the website more realistic, we also require:

1. Dashboard showing active machines, ongoing jobs, payments, and maintenance alerts.
2. Machine availability calendar.

3. Customer booking history.
4. Admin approval workflow for bookings.
5. Document upload for NIC, licence, and machine ownership proof.
6. Mobile-responsive design.
7. Audit logs for major actions performed by admins.

These responses can be used as your official client requirements and should provide enough information to design the complete UI, database, and business workflow for the system.

Source document: [Clarification_Questions.pdf \[Clarificat..._Questions | PDF\]](#)

additional things bot said

Yes. If I were a real client who had worked in the construction equipment industry for many years, there are some "hidden requirements" that customers often forget to mention during interviews. These usually appear later and cause change requests.

16. Site Visit Requests Before Booking

Sometimes customers do not know which machine they need.

We would like customers to request a site inspection before making a booking.

The system should allow:

- Site visit request
- Preferred date
- Project description
- Project location
- Photos of the work site

17. Fuel Responsibility Tracking

Different projects have different fuel arrangements.

The booking should specify:

- Fuel provided by customer
- Fuel provided by company
- Additional fuel charges

This avoids disputes later.

18. Overtime Tracking

Sometimes work continues beyond the agreed booking period.

We need:

- Planned end date
 - Actual end date
 - Overtime days
 - Additional charges
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19. Machine Replacement Workflow

If a booked machine becomes unavailable:

- Admin can assign a replacement machine.
 - Customer must be notified.
 - System keeps a history of the replacement.
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20. High-Priority Customers

Some customers are regular clients who bring significant business.

The system should mark:

- VIP customers
- Corporate customers
- Government customers

These customers may receive priority during machine allocation.

21. Corporate Customer Accounts

Construction companies often have multiple employees making bookings.

One company account should allow:

- Multiple staff members
 - One billing contact
 - One company profile
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22. Project-Based Bookings

Customers often hire machines for projects rather than individual jobs.

Each booking should be linked to:

- Project name
 - Site location
 - Project manager
 - Expected completion date
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23. Contract Document Storage

We want to keep copies of:

- Rental agreements
- Customer approvals
- Ownership documents
- Insurance documents

Everything should be easy to download later.

24. Machine Utilization Monitoring

Management wants to know:

- Which machine earns the most income
- Which machine stays idle
- Which machine has high maintenance costs

This helps future purchasing decisions.

25. Return Inspection Report

When a machine returns:

- Inspection notes
- Damage report
- Photos
- Responsible party

should be recorded.

26. Seasonal Demand Analysis

Although not needed immediately, the system should store enough data to identify:

- Busy months
 - Slow months
 - Popular machine categories
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27. Supplier Management

We buy spare parts from many suppliers.

The system could store:

- Supplier information
 - Contact details
 - Parts purchased
 - Payment records
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28. Internal Messaging

Sometimes operators, mechanics, and office staff need to communicate.

Instead of using external apps, a simple internal note system would be useful.

29. Safety Compliance Records

For each machine we should record:

- Safety inspections
- Compliance certificates
- Inspection dates
- Expiry dates

Construction companies often ask for these records.

30. Digital Signature Support

Customers should be able to digitally acknowledge:

- Booking confirmation
- Equipment handover
- Equipment return

This reduces paperwork.

31. Equipment Accessories Management

Some machines are hired with extra attachments.

Examples:

- Buckets
- Breakers
- Grapples
- Fork attachments

The system should track these separately.

32. Expected Revenue Forecast

Management would like a dashboard showing:

- Upcoming bookings
 - Estimated future revenue
 - Expected machine utilization
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33. Employee Leave Impact

If an operator is on leave:

- The operator should not be assignable.
 - The system should suggest another available operator.
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34. Customer Documents Archive

Customers often repeatedly upload the same documents.

The system should keep:

- NIC copies
- Company registration documents
- Other approvals

so they can be reused.

35. Business Rules Often Missed by Students

As a client, I would specifically tell you:

- A machine cannot be booked while under maintenance.
- An operator cannot be assigned to two jobs on the same date.
- Every completed booking must generate an invoice.
- Every payment must have a receipt.
- Every maintenance record must have a responsible mechanic.
- Every machine must belong to either the company or a private owner.

- Cancelled bookings should remain in history and not be deleted.
 - Deleted records should be archived for auditing.
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Features That Would Impress Your Lecturer

If I were your client, these would make the project look very professional:

1. QR code for each machine.
2. Booking approval workflow.
3. Digital document storage.
4. Complaint management.
5. Damage tracking with photos.
6. Machine replacement process.
7. Revenue analytics dashboard.
8. VIP customer management.
9. Contract management.
10. Audit log/history tracking.

These are the kinds of requirements that real clients often reveal only after several meetings, and including them will make your SRS, use cases, database design, and website UI feel much more realistic and industry-level.

Yes, as the client, I would want **most of those features** in the system. However, not all features have the same priority.

Must Have (Include in Final System)

- User Management
- Machine/JCB Management
- Booking Management
- Customer Management
- Operator Management
- Maintenance Management
- Payment Management
- Invoice Generation
- Notifications
- Reports & Analytics
- Dashboard
- Machine Availability Calendar
- Document Uploads
- Complaint Management

- Customer Ratings & Reviews
- Blacklist Customers
- Audit Logs
- Machine Ownership Tracking
- Commission Tracking
- Expiry Alerts
- Damage Tracking
- Booking Approval Workflow
- Return Inspection Reports

Should Have (Good Features)

- Overtime Tracking
- Site Visit Requests
- Fuel Responsibility Tracking
- Project-Based Bookings
- Contract Document Storage
- Corporate Customer Accounts
- VIP Customer Management
- Machine Replacement Workflow
- Employee Leave Management
- Revenue Forecast Dashboard
- Equipment Attachment Management
- QR Code for Machines

Can Be Added Later (Future Enhancements)

- GPS Tracking
- Mobile Application
- Online Payment Gateway
- SMS Notifications
- Internal Messaging
- Supplier Management
- Seasonal Demand Analysis
- AI-Based Machine Recommendations
- Digital Signature Integration

Not Needed for This Version

- Government NIC Verification APIs
- Live Vehicle Tracking Hardware Integration
- Advanced Accounting System

- Payroll Processing for Employees
- Complex ERP Features

If I Were the Client Signing Off the Requirements

I would say:

"For the first release, I want the system to focus on machine bookings, operators, maintenance, payments, invoices, customer management, reports, and dashboards. Additional features such as GPS tracking, mobile apps, and AI recommendations can be developed in future phases."

For a university SE project, I recommend implementing about **15-20 core features** rather than trying to build all 35+ features. That will make the project realistic, manageable, and professional.